

Smart Campus Navigator

Revolutionizing the Campus Experience

.

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The Challenge of Campus Life

The Challenge:

- **Large Campus Navigation:** Thapar Institute spans a vast area with 100+ buildings
- **New Student Orientation:** Difficulty finding classrooms, labs, and facilities
- **Event Discovery:** Limited awareness of campus events and their locations
- **Visitor Assistance:** External visitors struggle with campus navigation
- **Real-time Information Gap:** No centralized platform for locationbased services

User Need Analysis

(Based on Google Form)

To understand whether a campus navigation system is genuinely needed, a Google Form–based survey was conducted among students across different years. The survey focused on difficulties faced while navigating the campus, pain points, and expectations from a digital navigation tool.

Key Pain Points Identified

Across responses, the most common navigation problems were:

- Difficulty finding labs, departments, classrooms, and event venues
- Lack of a centralized digital campus map
- Inability to quickly locate events or competitions
- Confusing building layouts and multiple floors
- No dedicated platform showing walking routes
- Reliance on friends or random directions

High Interest in a Campus-Specific Map

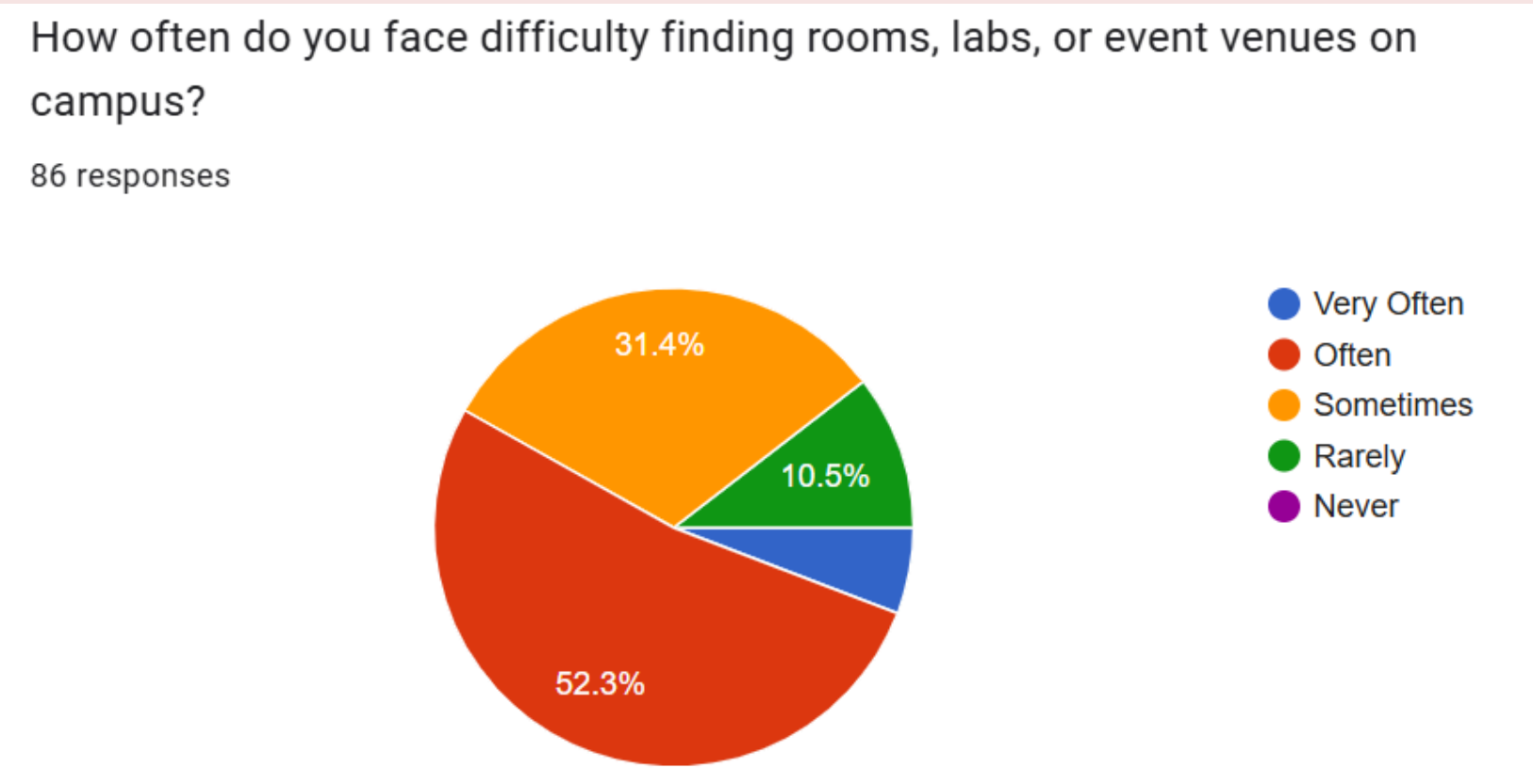
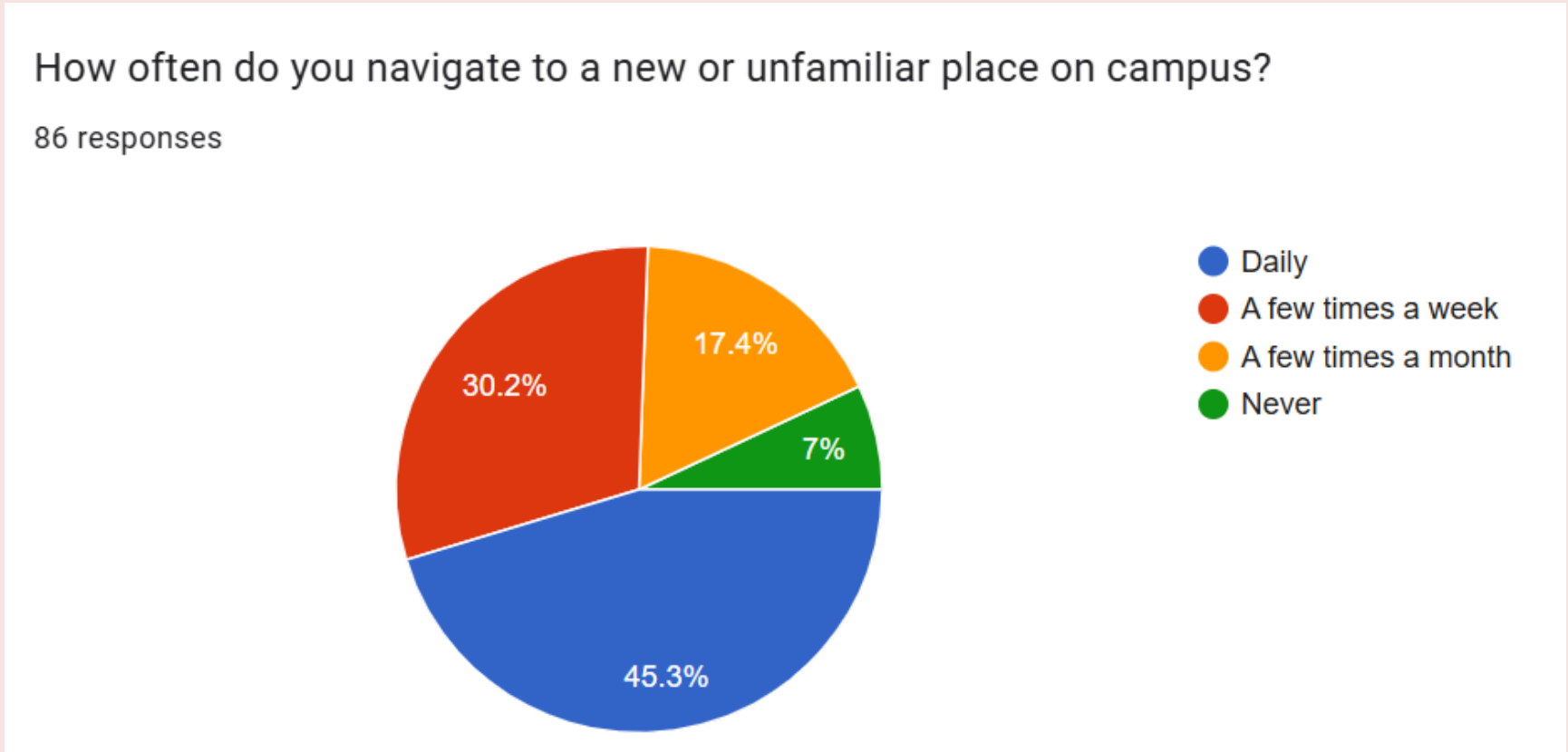
Most students rated the usefulness of a campus map with walking routes as 4 or 5 out of 5, demonstrating strong demand for such a tool.

High Likelihood of Adoption

When asked how likely they would be to use such an app if it were available:

- A majority responded “Likely” or “Very Likely”
- Very few were neutral or unlikely

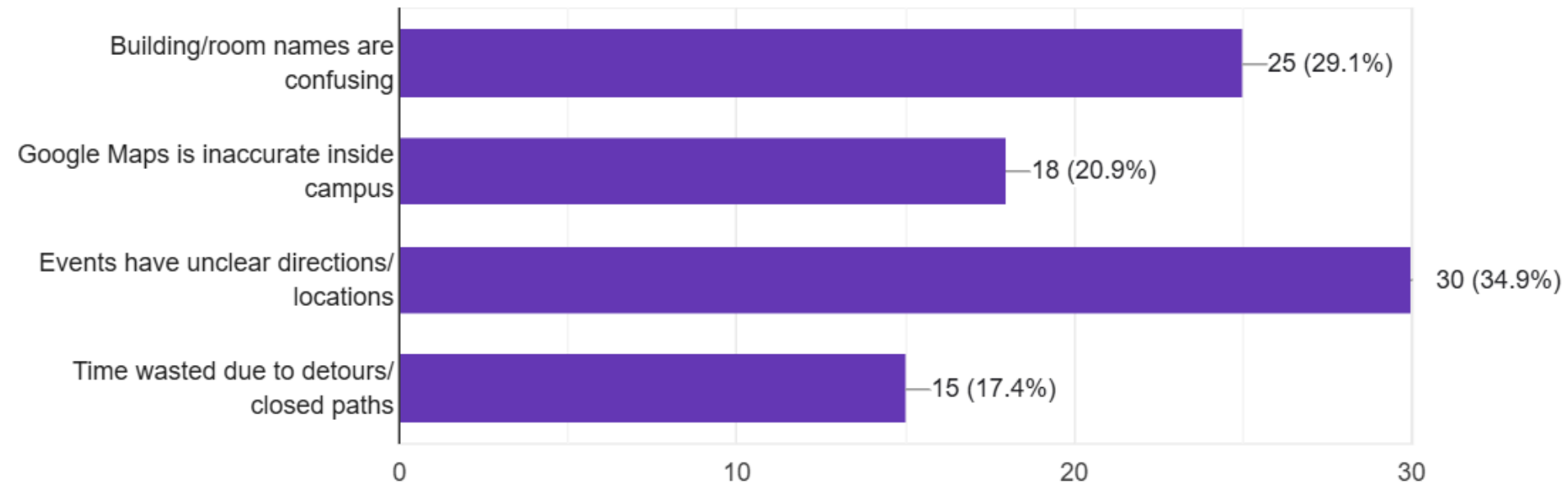
Here are some snapshots regarding the same



What are the top pain points you face while navigating?

 [Copy chart](#)

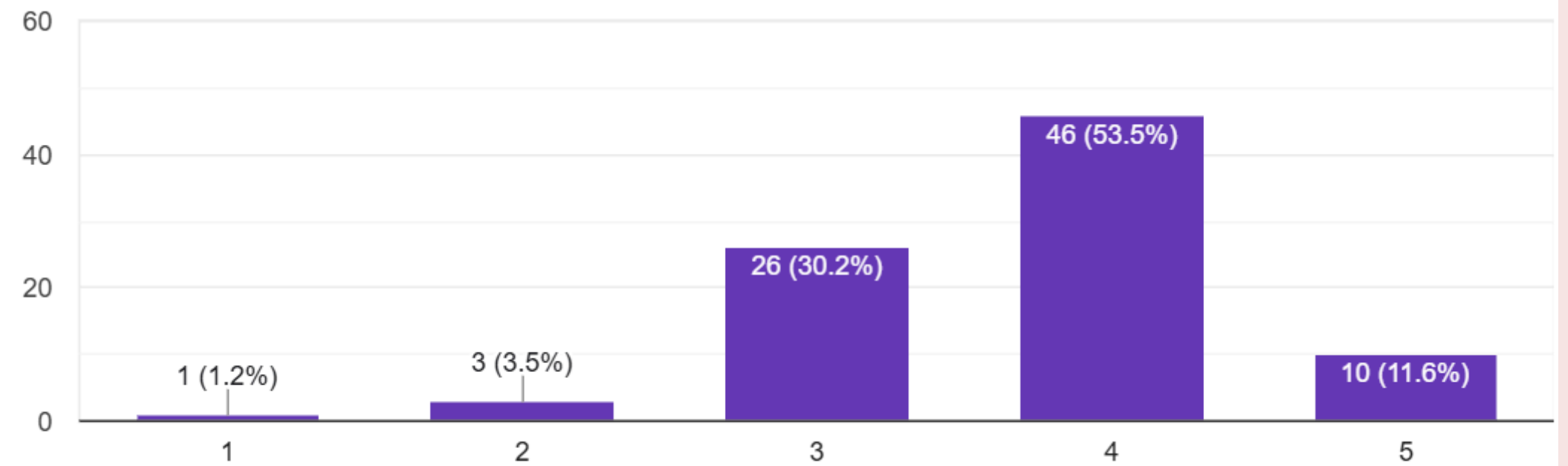
86 responses



How useful would a campus-specific map with walking routes be for you?

 [Copy chart](#)

86 responses



One Website for Everything Campus

A comprehensive website that integrates smart navigation and event discovery into an intuitive, browser-based experience.

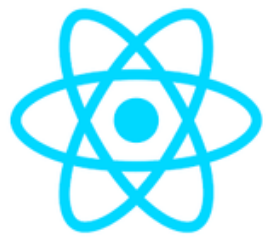
Key Features:

- **Interactive 2D Map:** A detailed, zoomable map of the entire campus, accessible on any device with a web browser.
- **Outdoor Navigation:** An accurate pathway with proper directions to any building.
- **Centralized Event Hub:** An up-to-date listing of all campus events.

Technology

Built on Modern Web Tech, Aiming for the Future

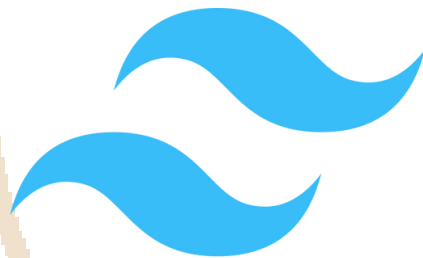
Frontend



React



Leaflet Maps



Tailwind CSS



Vite

Vite

Backend



JavaScript



ExpressJS

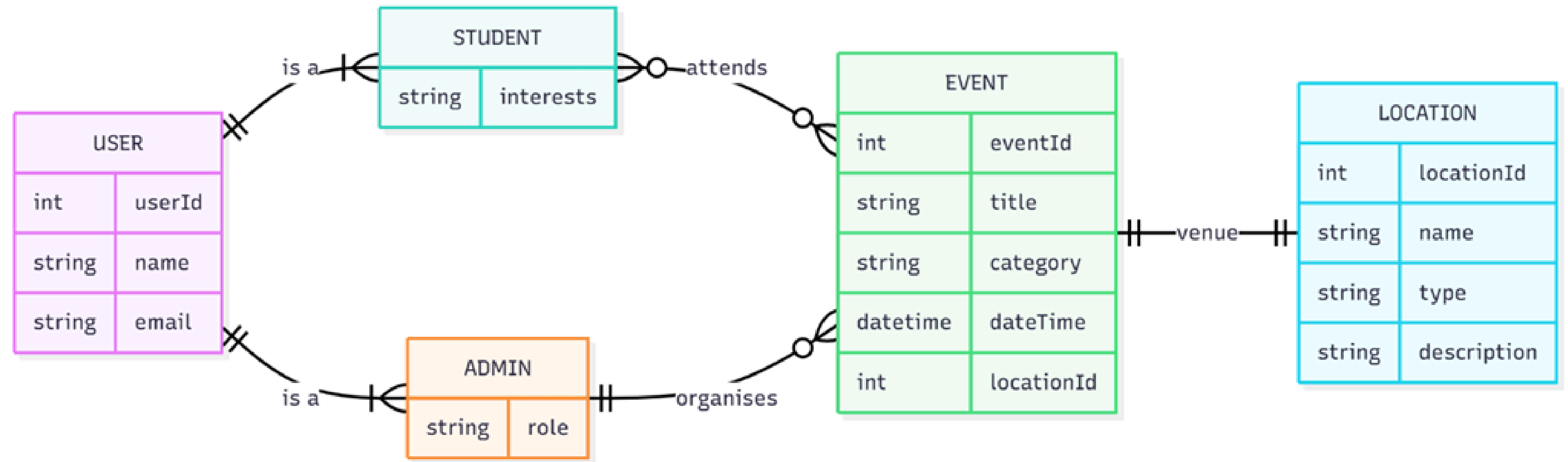


MongoDB



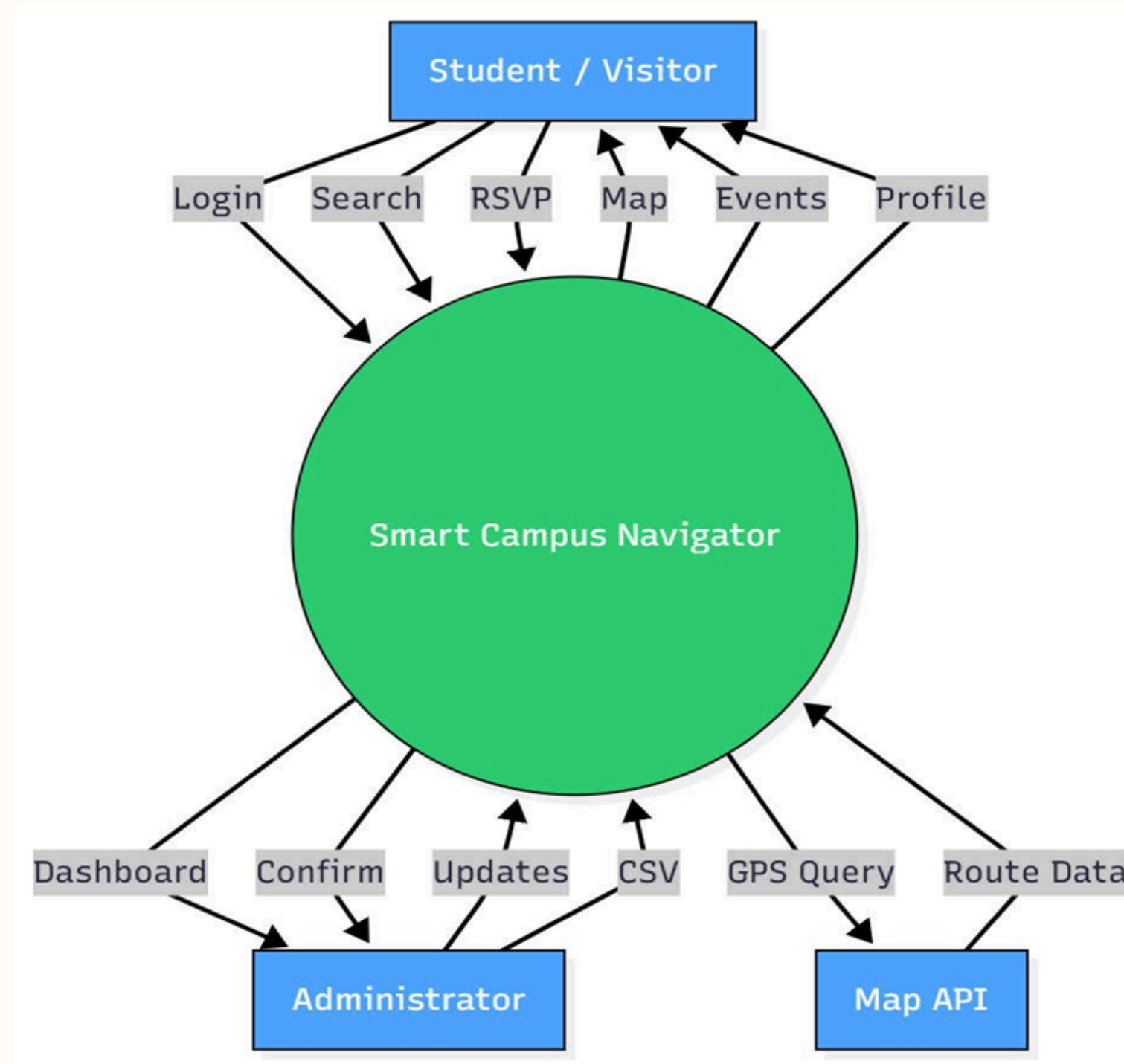
NodeJS

Design



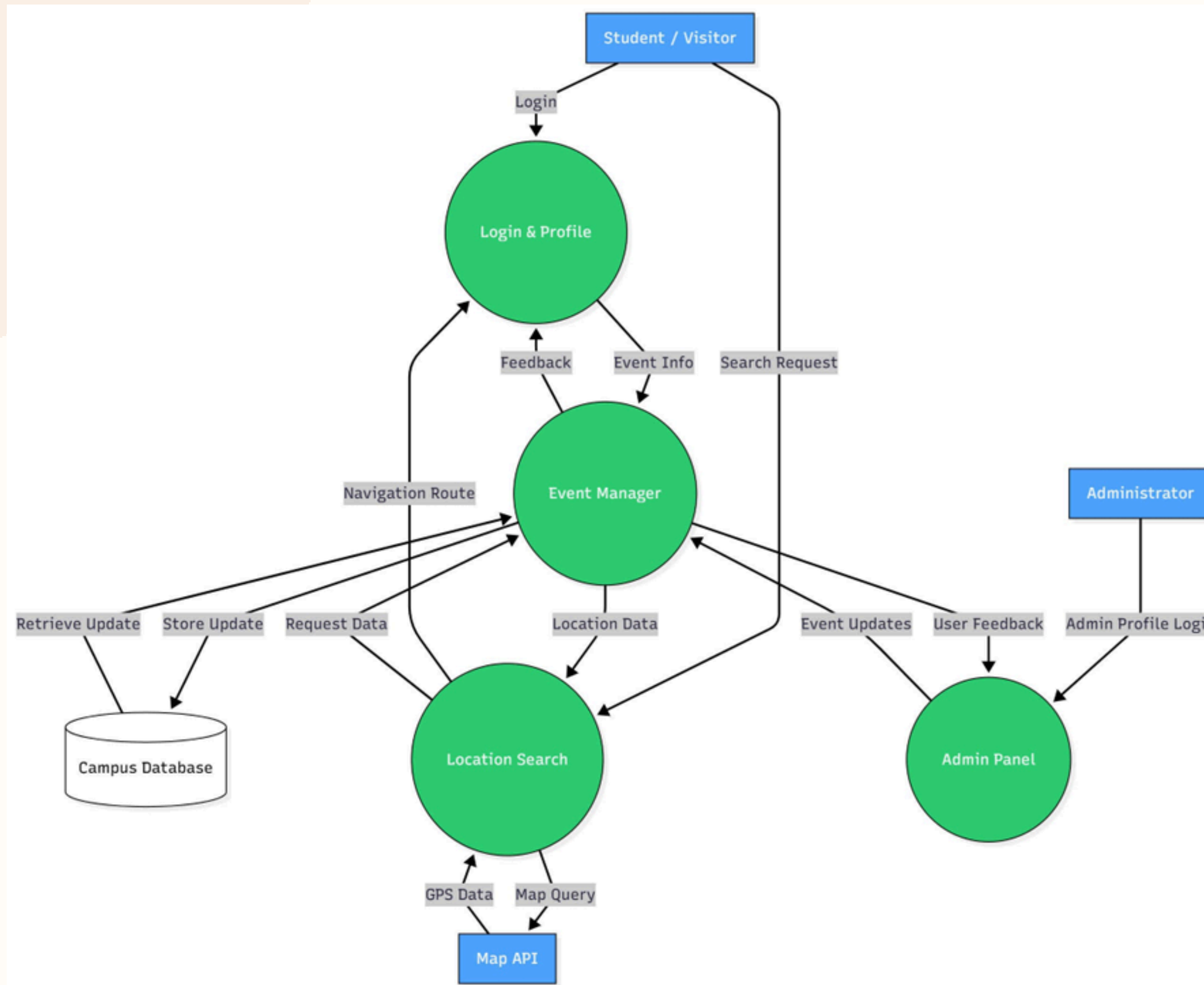
Entity-Relationship Diagram- Shows database entities and relationships.

Design contd.



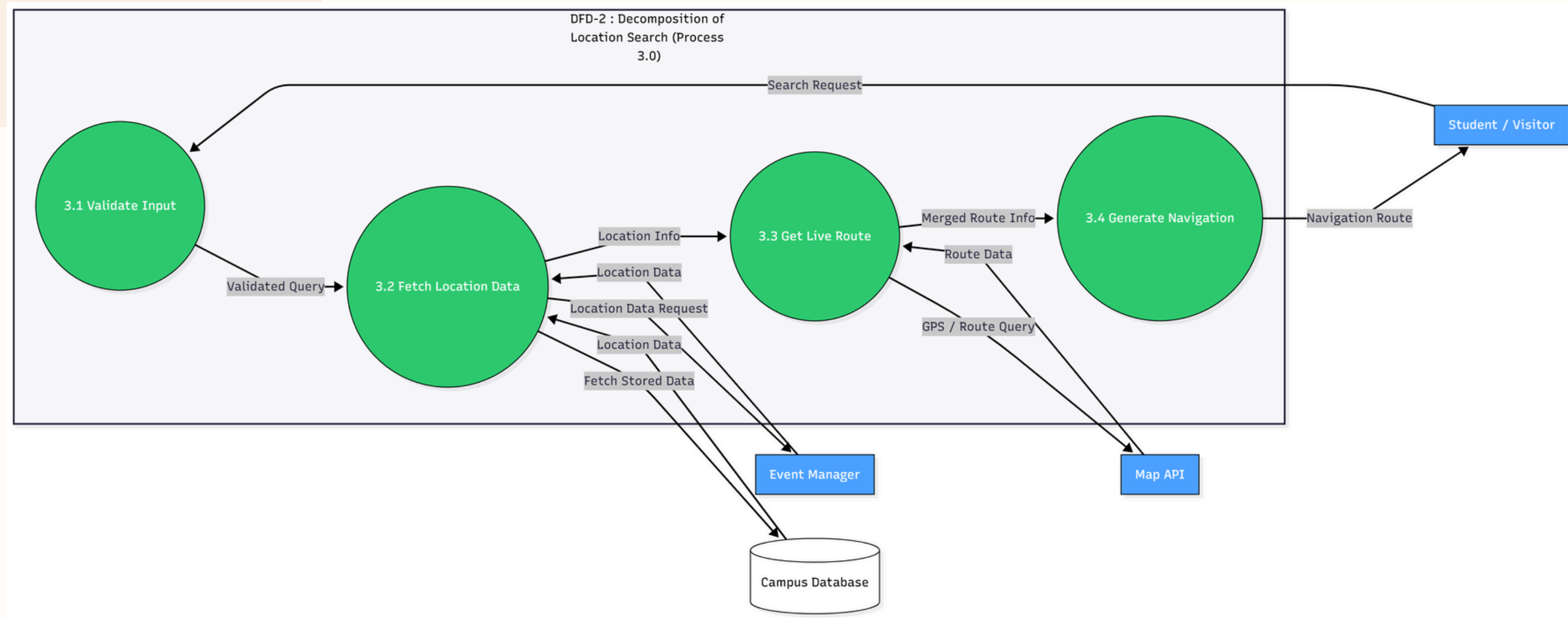
Context DFD (Level 0)– Shows main data flow between users and system modules.

Design contd.



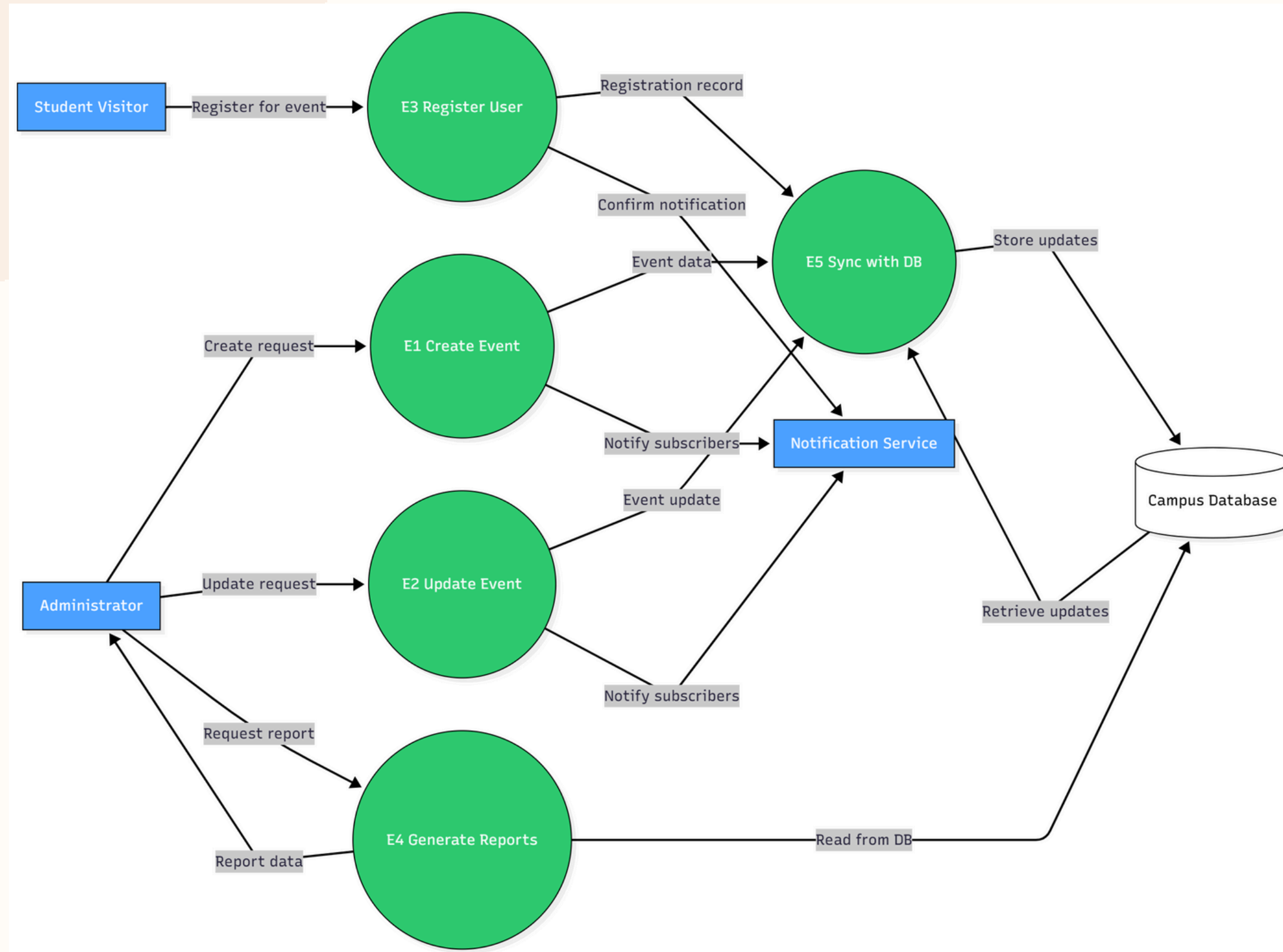
DFD Level 1– Detailed process flows of the system.

Design contd.



DFD Level 2 (Location Search) – Breakdown of the location search process.

Design contd.



DFD Level 2 (Event Management)– Breakdown of the event management process.

Core Functionalities

(Depicted using the deployed Model)

User Authentication

- Secure registration/login with JWT tokens
- Google Sign-In: Users log in with their Google account.



Welcome back

Sign in to Smart Navigator
Thapar Institute of Engineering & Technology

Email Address *

Password *

☐ Remember me [Forgot your password?](#)

Sign in

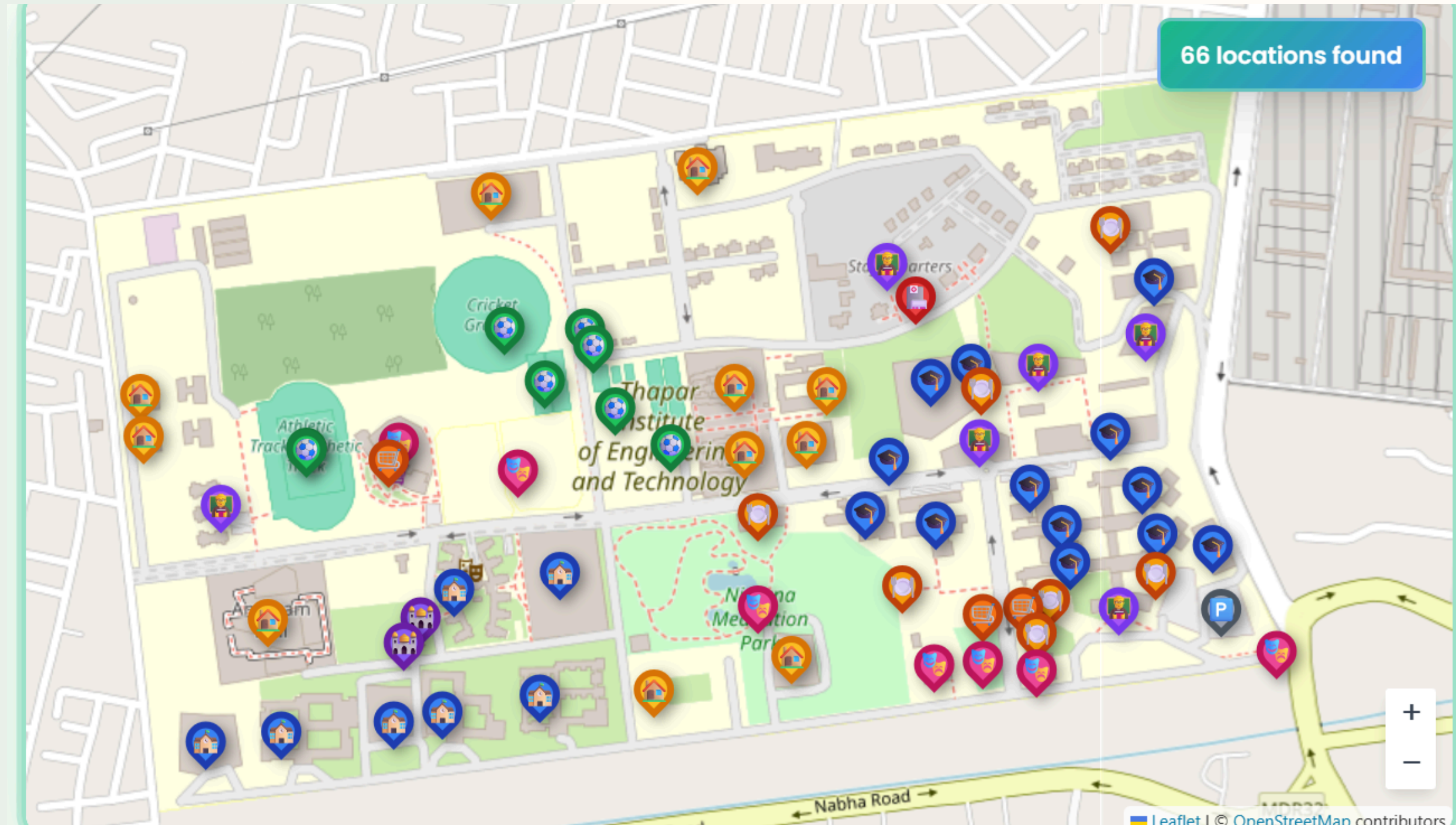
Or continue with

 Continue with Google

Don't have an account? [Sign up here](#)

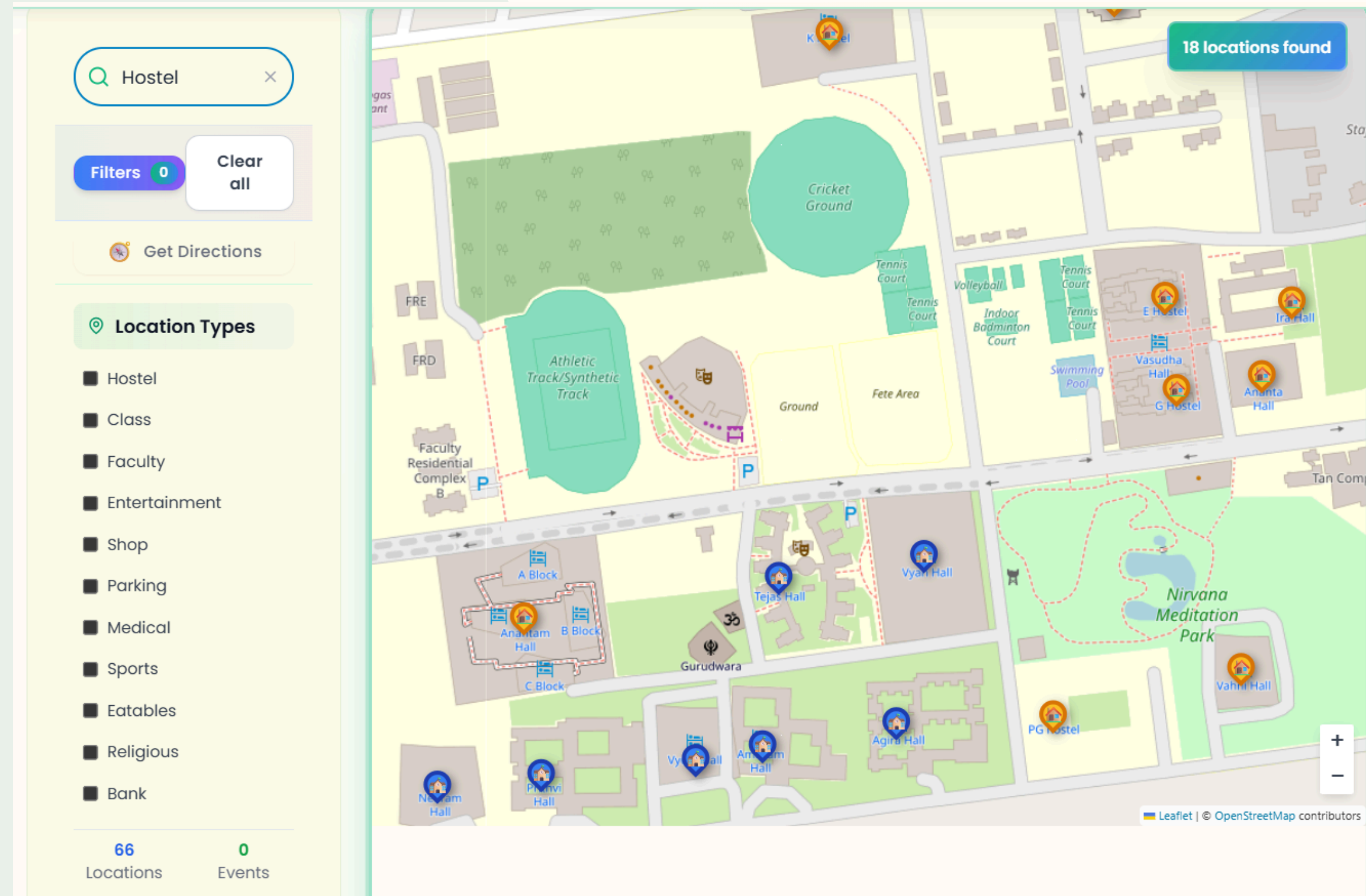
Interactive Campus Mapping

- Leaflet-based mapping with custom markers
- Real-time route planning between locations



Advanced Search & Filtering

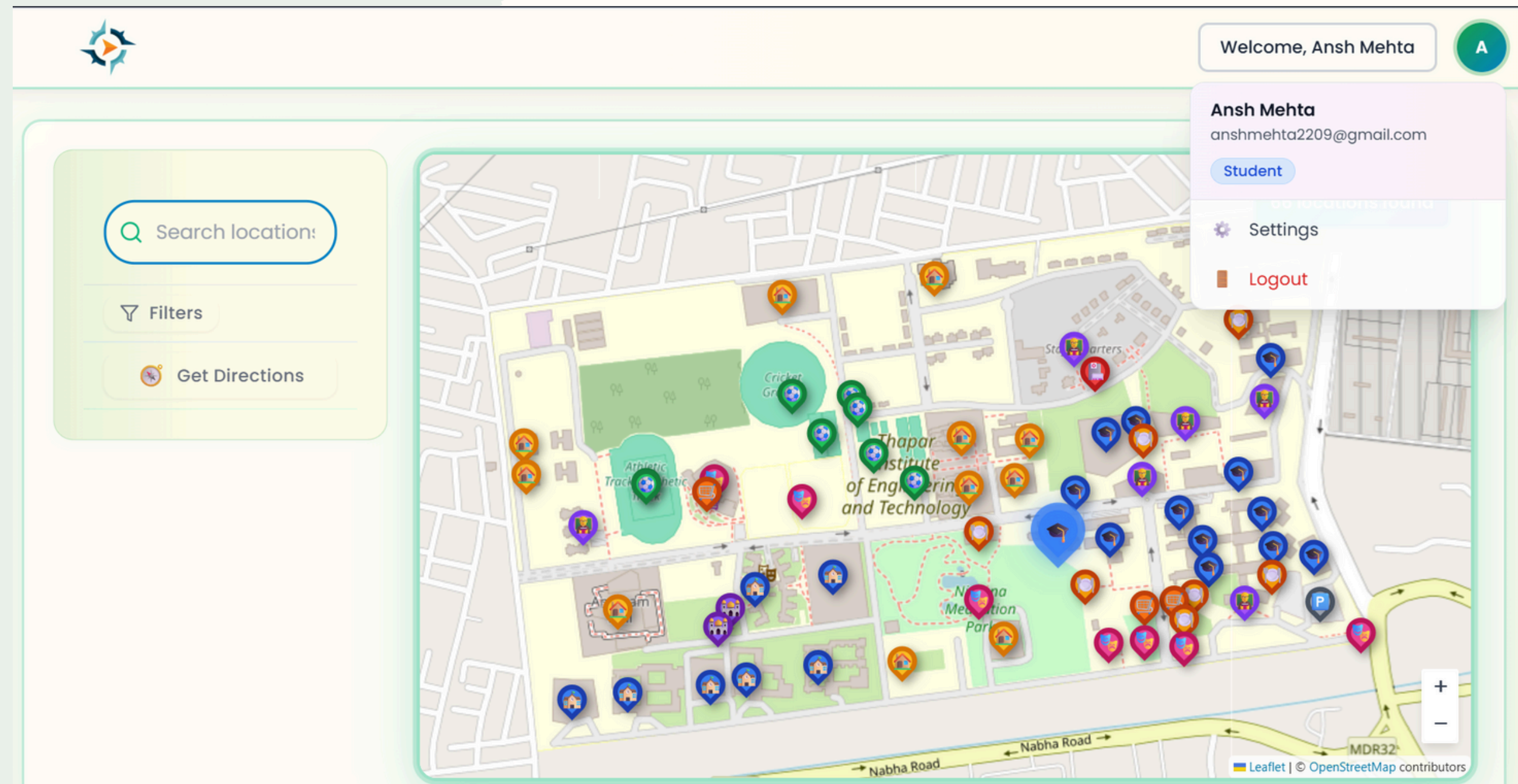
- Text-based search across locations and events
- Category filtering (Academic, Recreation, Dining)



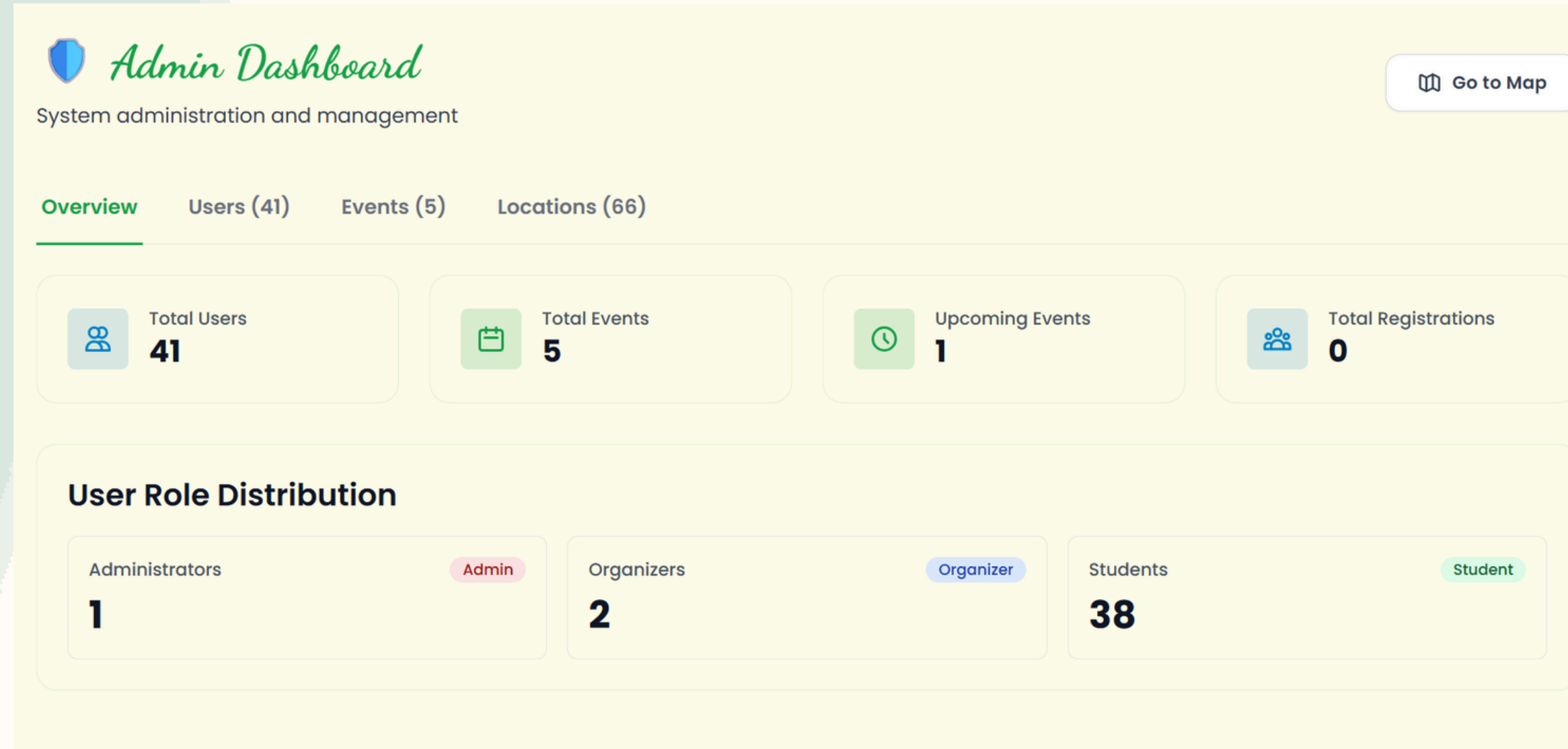
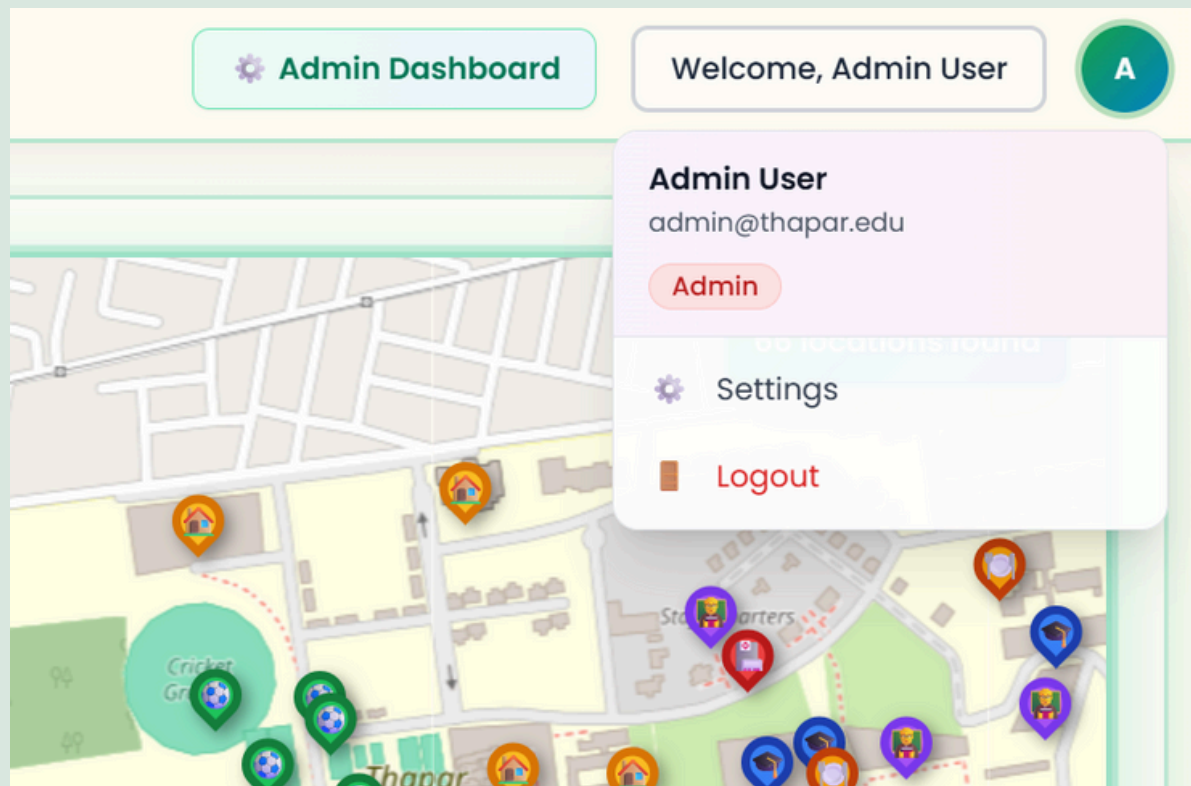
User Authorization

- Role-based access control (Student/Organizer/Admin)

Student

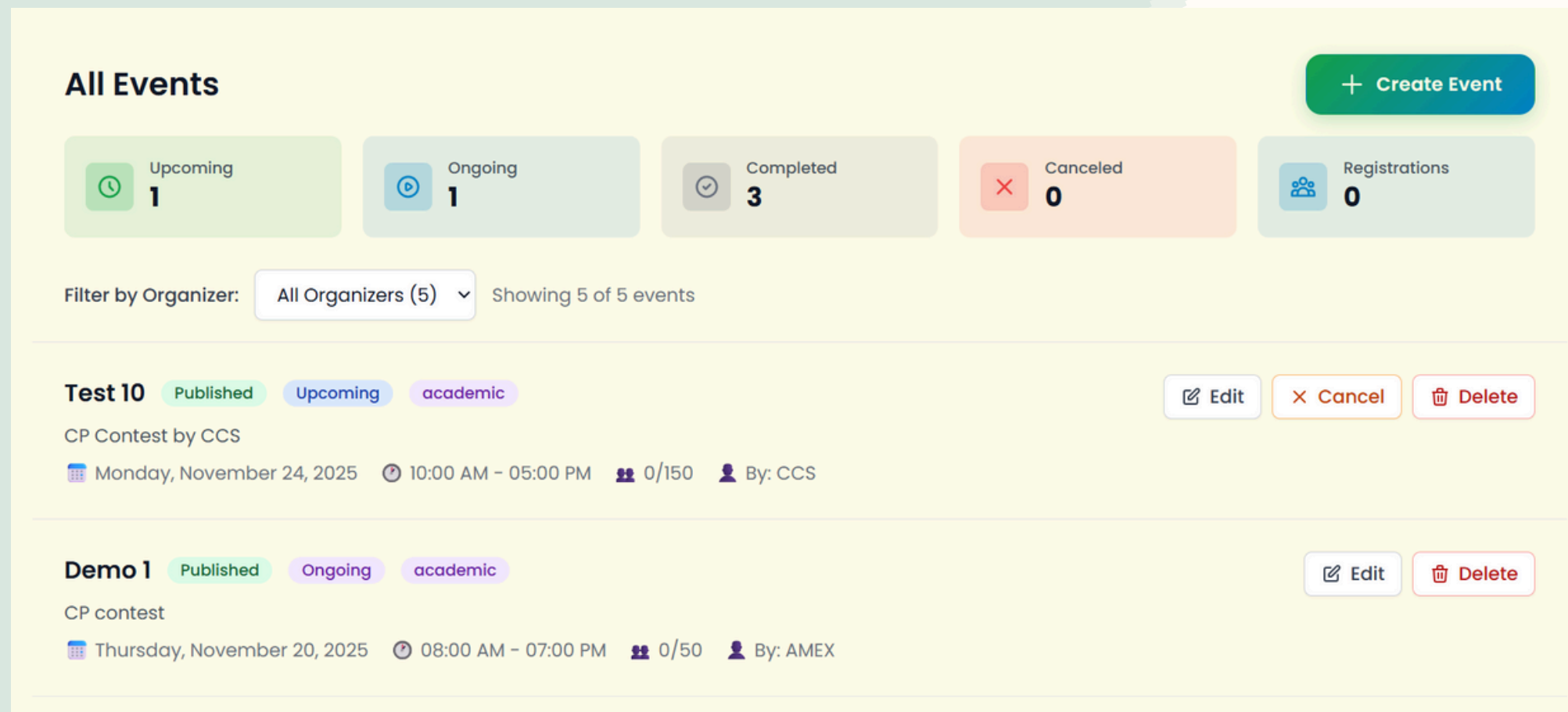


Admin

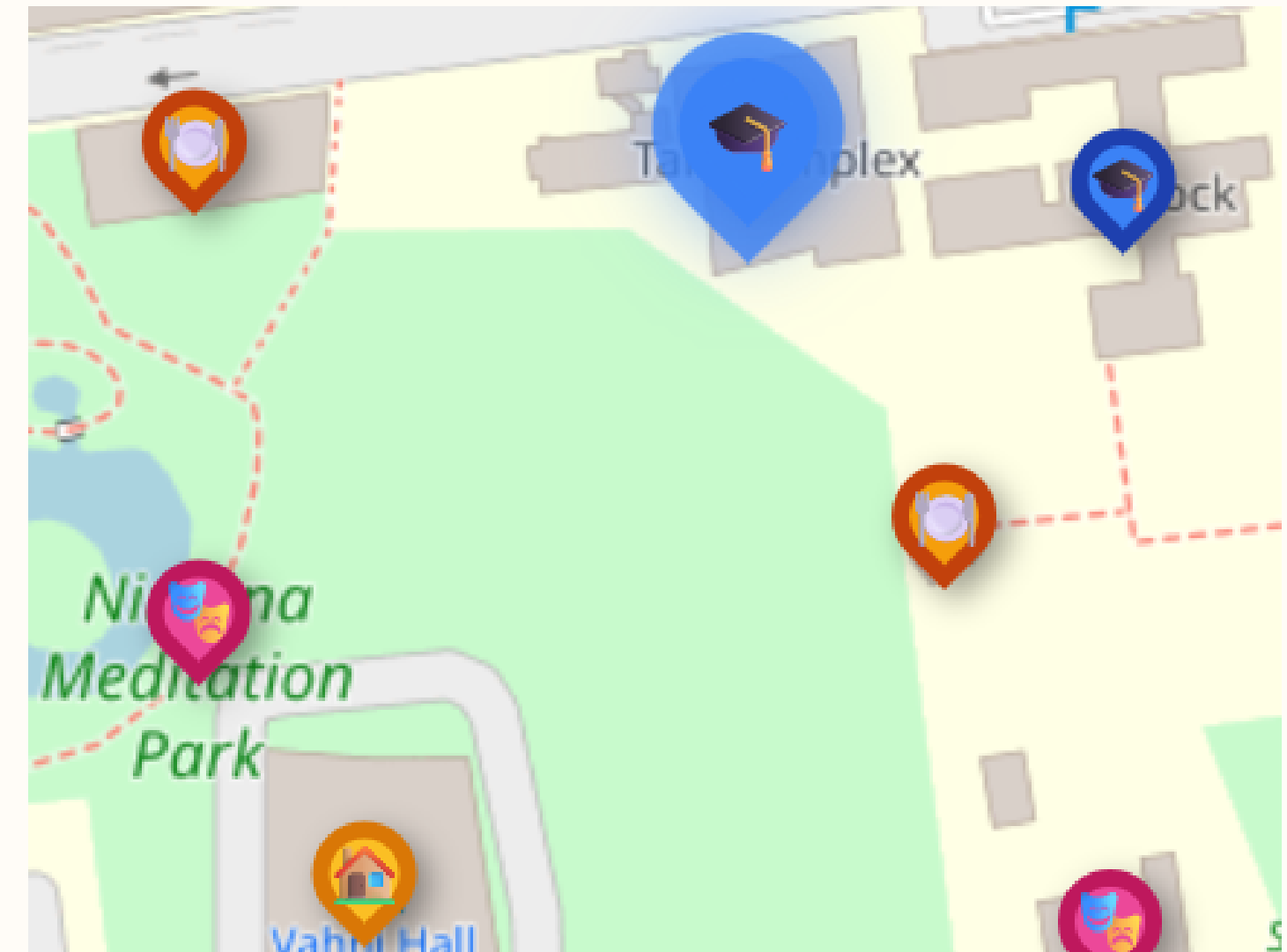


Event Hub

An up-to-date listing of all campus events
Add/Remove Events.



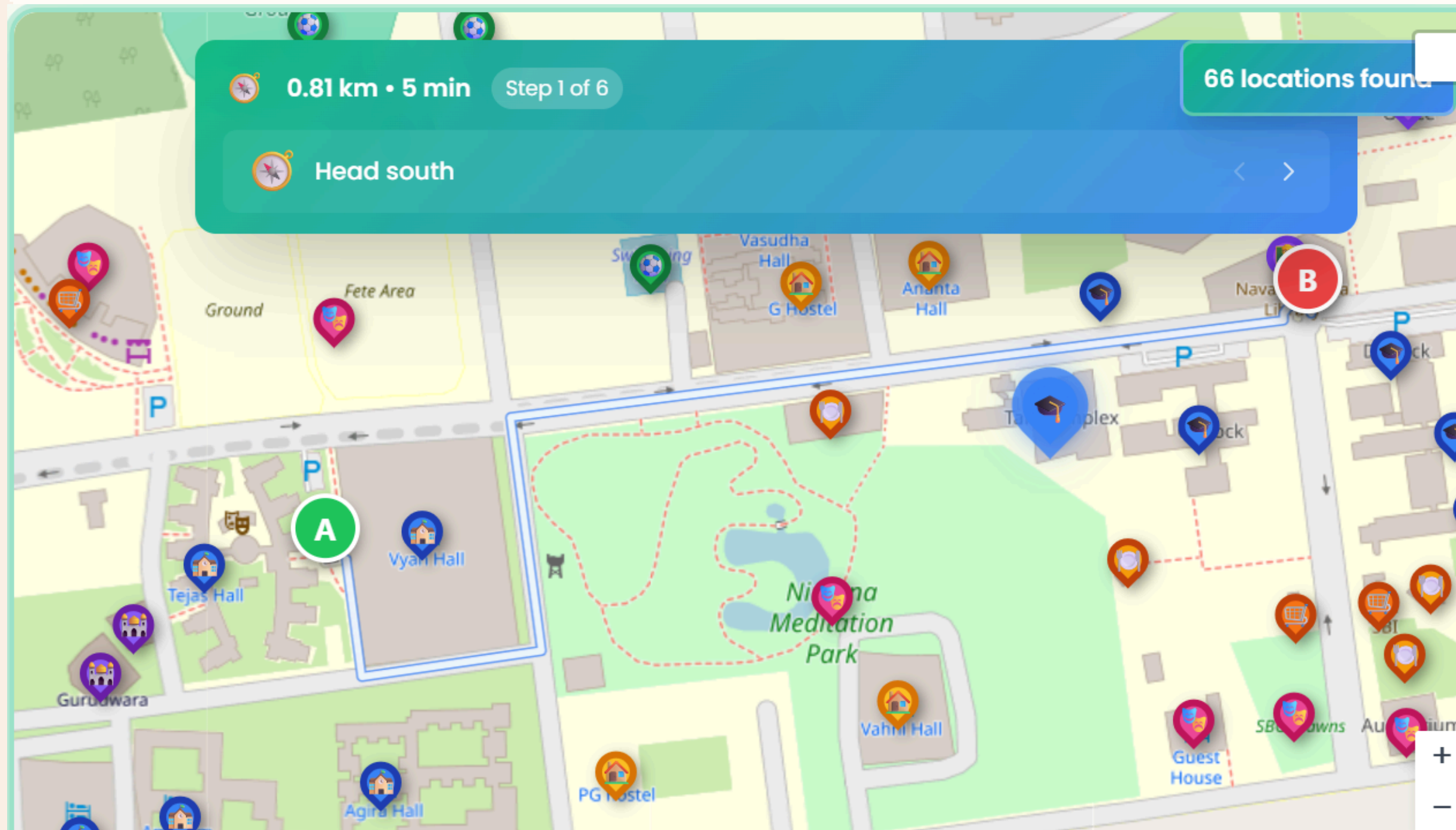
Admin/Organizer view for event management



Event marked by highlighting the location pin of the upcoming event

Path Finder

- **Point & Navigate:** Simply click two locations to generate an optimized walking route with distance and time estimates.
- **Clear Visual Guidance:** Distinctive markers and route lines provide instant visual confirmation of your planned path.
- **Detailed Instructions:** Collapsible directions panel offers step-by-step navigation guidance for seamless campus traversal.



Future Implications

(Expanding SmartNav beyond Outdoor Use)

SmartNav can evolve into a complete navigation ecosystem that works seamlessly both outdoors and indoors. Key upcoming directions include:

- **Indoor Navigation**
- **Live Step-by-Step Navigation**
- **Advanced Routing & Real-Time Context Awareness**

Indoor Navigation

Why Indoor Navigation?

- GPS signals are weak or unavailable inside buildings
- Large campuses, malls, hospitals, and airports need micro-level guidance
- Reduces confusion, time-wastage, and dependency on signboards

Future Directions

- Using Bluetooth Beacons, Wi-Fi Fingerprinting, or UWB signals for indoor positioning
- Creating detailed floor-level maps and room-to-room routes
- SmartNav app auto-switches between outdoor GPS and indoor sensors
- Integration with emergency evacuation paths

Live Step-by-Step Navigation

What It Adds

- Users receive real-time instructions as they move
- Simple tap-based directions:
- “Turn left”, “Go straight for 20m”, “Take the stairs to Floor 2”

Future Enhancements

- AR-based arrows showing direction in real-time
- Voice-guided indoor and outdoor navigation
- Re-routing dynamically if the user deviates from path
- Step-wise vibration alerts for accessibility

Why Step-by-Step Navigation Is Not Yet Implemented

1. Requires Real-Time Location Tracking

Step-by-step navigation depends on continuous GPS/position updates.

But in our current prototype:

- We capture the user's location only once to generate the route.
- No background service continuously monitors user movement.

2. “No free API currently supports complete step-by-step navigation. Only basic routing is available in free tiers, while live navigation is offered under paid SDKs.”

Even the free Google Maps JS API does NOT provide:

- Turn-by-turn navigation
- Real-time navigation UI
- Route progress tracking

Documentation By Google



Feasibility Analysis

Technical Feasibility:

Aspect	Assessment	Notes
Stack Maturity	High	React, Express, MongoDB are battle-tested
Geospatial Queries	Feasible	MongoDB 2dsphere + indexes already functioning
Real Time Extension	Moderate	Could add WebSockets or SSE later
Offline/PWA	Feasible	Requires service worker + caching strategy
Scaling	Horizontally Scalable	Stateless JWT, need external session revocation strategy if added

Economical Feasibility:

Cost Element	Current(Dev)	Future(Pilot)
Hosting (App + DB)	Free tier / minimal	~\$30–50/mo (cloud VM + managed DB)
Domain & SSLFeasible	Dev placeholder	~\$10–15/yr domain + managed cert
Monitoring / Logs	Minimal	Add low-tier APM / log storage
Developer Effort	Student hours	Additional 4–6 person-weeks for roadmap phase

Operational Feasibility:

Factor	Status	Action
Admin Workflow	CRUD UI functional	Add moderation dashboard
Data Maintenance	Manual seeding	Build import + update scripts
Support & Training	Low complexity UI	Provide quickstart guide

User Evaluation

(Based on Google Form)

To evaluate the usability and performance of the SmartNav system, a Google Form–based feedback survey was conducted among beta testers. The participants interacted with the deployed website and provided responses regarding layout intuitiveness, responsiveness, functionality, and overall user experience.

Key Findings:

- **Navigation Layout:**

- On a scale of 1–5, the average rating for layout intuitiveness was 4.2, indicating that users found the interface clear and easy to understand.

- **Ease of Use:**

- Most users reported being able to find the desired information easily without confusion or unnecessary clicks.

- **Responsiveness:**

- The majority mentioned that SmartNav adapted well to different screen sizes, with only minor alignment issues on mobile for some users.

Key Findings:

- **Performance:**

4 out of 5 users reported no lag or delay during interaction, showing smooth transitions and animations.

- **Navigation Functionality:**

The “active page” indicator worked correctly for most testers, ensuring good orientation while navigating through the website.

- **Accessibility:**

Around 60% of users could navigate using the keyboard (Tab/Enter), indicating partial accessibility support.

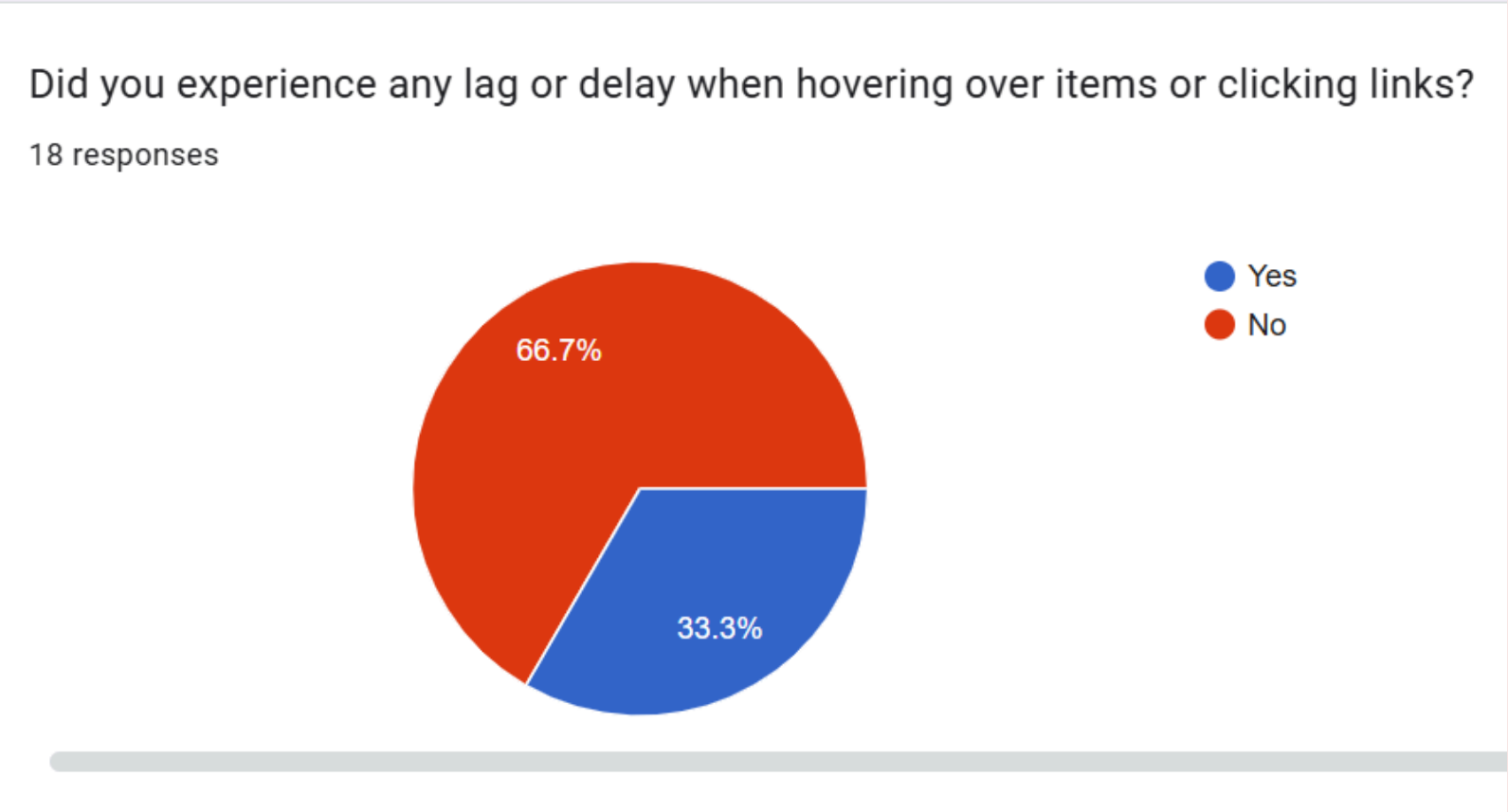
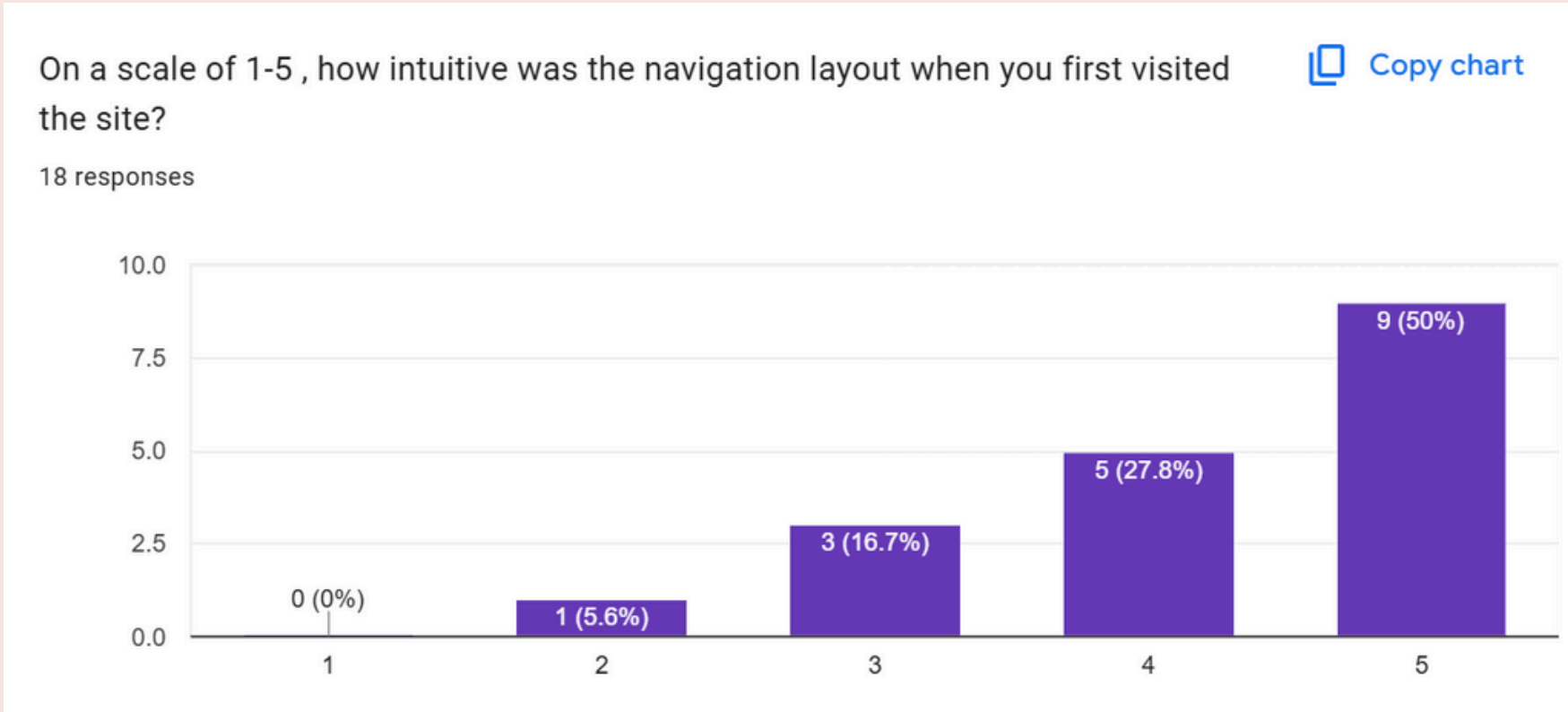
- **Readability:**

All users agreed that text contrast was sufficient for comfortable reading.

- **Bugs:**

No major bugs were encountered; minor observations were noted but not reproducible.

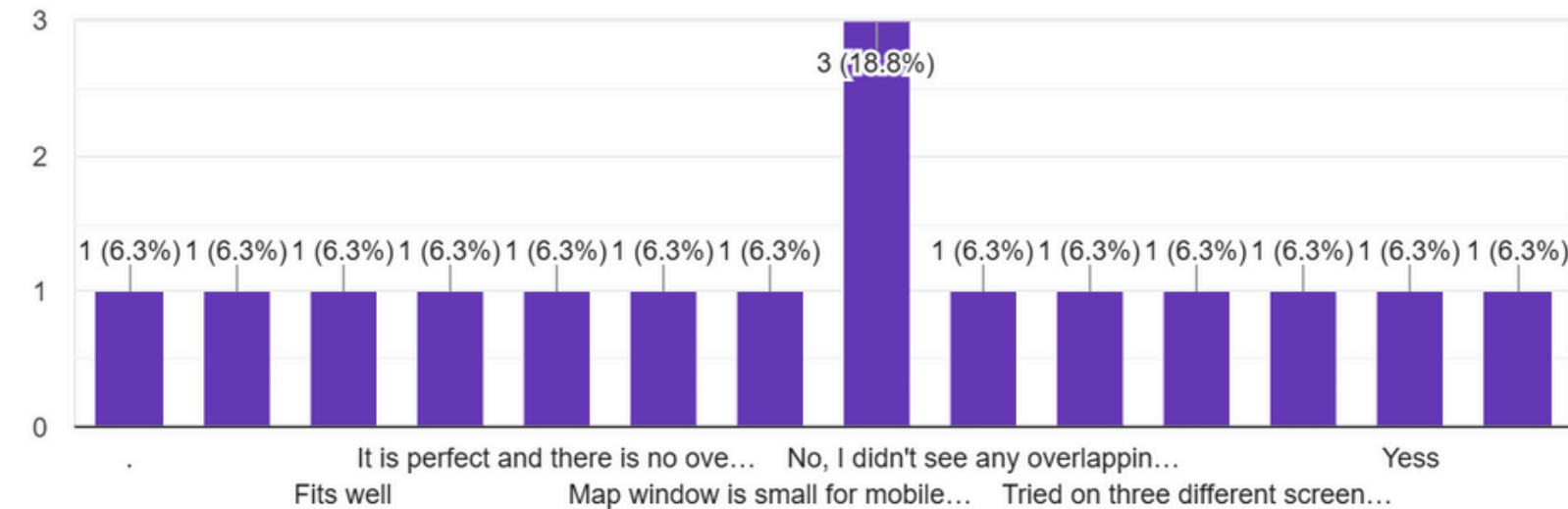
Here are some snapshots regarding the same



How well did the SmartNav adapt to your screen size? Did you notice any overlapping text , cut-off elements , or misalignment on mobile vs desktop?

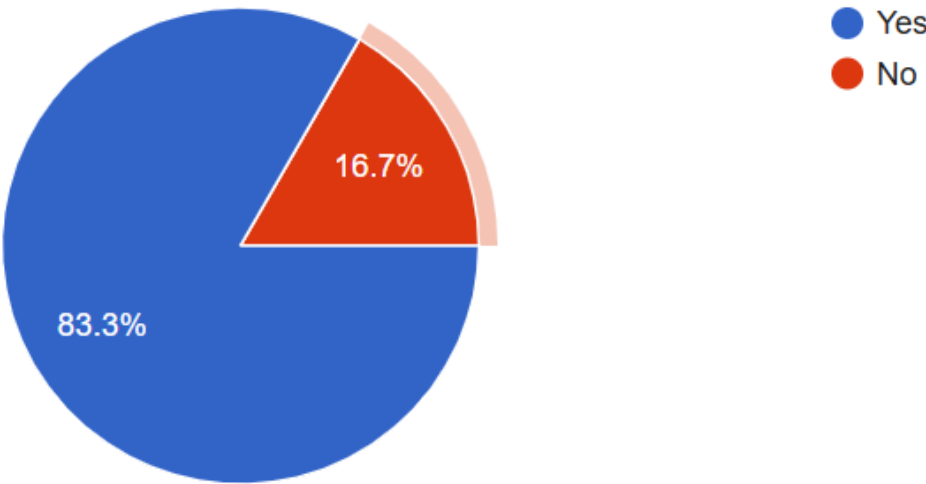
 [Copy chart](#)

16 responses



Was the text contrast sufficient for reading against the background color of the navigation bar?

18 responses



Impact And Benefits

For Students

- 50% reduction in time spent finding locations
- Increased event participation through better discovery
- Enhanced campus experience with personalized features

For Faculty & Staff

- Streamlined event management with automated tools
- Better resource allocation through usage analytics
- Improved visitor assistance capabilities

For Institution

- Enhanced digital presence and modern image
- Improved student satisfaction scores
- Better space utilization through analytics
- Cost savings in physical signage and maps

Key TakeAways

- ***Solves the problem of campus navigation and event discovery***
- ***Built with a modern and scalable MERN stack architecture***
- ***Implements strong security with JWT authentication and RBAC***
- ***Provides an interactive, responsive, and accessible user interface***
- ***Future-ready with plans for AR, mobile app, and IoT integration***

Scan to Visit the Website

SmartNav – Live and Online





Thank you!